Interpreting Linear Models

Revision sheet — Data Analysis for Psychology in R 2 (DAPR2), Week 2 · University of Edinburgh 2026–27

Learning objectives (= what can be examined)

- Be able to **interpret the coefficients** of a simple linear model.
- Understand **how and why we standardise** coefficients, and how this changes interpretation.
- Understand how interpretation changes when we **add more predictors** (multiple regression).

1. Recap

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

y = outcome, x = predictor, β_0 = intercept, β_1 = slope, ε = residual (observed − predicted). In R: `lm(DV ~ IV, data = datasetName)`. Running example: `lm(score ~ hours, data = test)`.

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.4000     1.1111   0.360   0.7282
hours         1.0545     0.3581   2.945   0.0186 *
Residual standard error: 1.626 on 8 degrees of freedom
Multiple R-squared: 0.5201,     Adjusted R-squared: 0.4601
F-statistic: 8.67 on 1 and 8 DF, p-value: 0.01858
```

2. Interpreting the coefficients

Coefficient	General definition	In the example
Intercept (β_0 = 0.40)	The **expected value of Y when X = 0**.	A student who does **not study** (hours = 0) is expected to score **0.40** on the test.
Slope (β_1 = 1.055)	How much Y increases, **on average**, when X increases by **one unit**.	For every **extra hour** of study, test score increases on average by **1.055 points**.

Exam template

Always state the answer **in the units of the variables**: unit of Y = 1 test point, unit of X = 1 hour. Template: "*For every one-[unit of X] increase in [X], [Y] changes on average by [β_1] [units of Y].*"

Caution: the intercept is not always meaningful

- Here X = 0 is meaningful (0 hours of study) — but **often it isn't**.
- If the predictor were **age**, an intercept would be the expected score at age 0: nonsensical.
- If a scale runs from 1 to 5, X = 0 lies **outside the possible range**.
- General lesson: **interpretation is always in the context of the constructs and how we measured them**.

3. Practice: scales of measurement (lecture exercises, with answers)

Model	Units	β_0 means...	β_1 means...
Salary (y) ~ duration of employment (x)	x: 1 year y: £1000 $\beta_1 = 0.4$	Expected salary (in £1000s) of an employee with **0 years** of employment, i.e. a new starter.	Each additional **year** of employment is associated with an average salary increase of **0.4 × £1000 = £400**.
Cat tail length (y) ~ weight (x)	x: 1 kg y: 1 cm $\beta_1 = -3.2$	Expected tail length of a cat weighing **0 kg** — not meaningful (no cat weighs 0 kg).	Each extra **kg** of weight is associated with a tail that is on average **3.2 cm shorter** (negative slope).
Healthy eating (y) ~ conscientiousness (x)	x: 1 point on a 1–5 Likert scale y: 1 point on healthy-eating scale $\beta_1 = 0.25$	Expected healthy-eating score at conscientiousness = 0 — **outside the 1–5 scale**, so not interpretable.	Each 1-point increase in conscientiousness is associated with an average **0.25-point** increase in healthy eating.

4. Standardisation

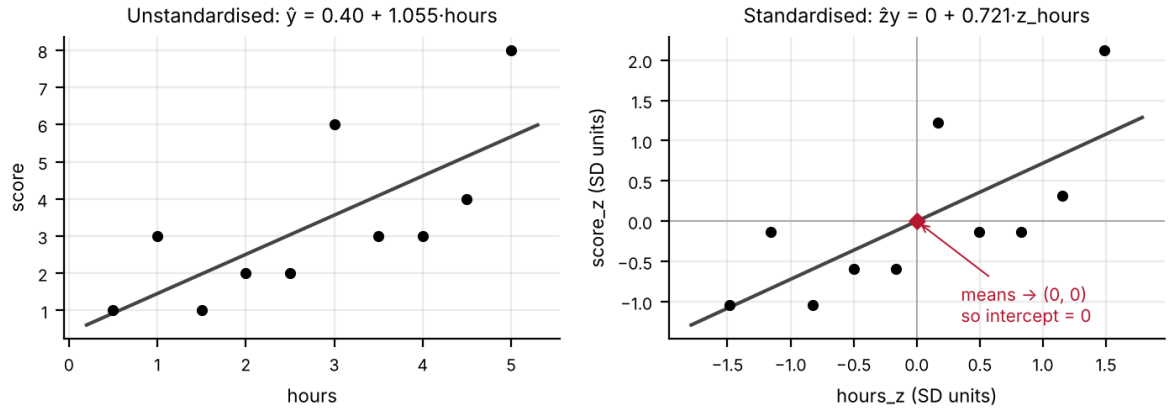
So far, coefficients were **unstandardised**: expressed in the original units of measurement. Sometimes those units are unhelpful, and we **standardise** to aid interpretation.

Why standardise?

- **Arbitrary scales**: e.g. a sum score of Likert items — "one unit" (moving from 2 to 3 on one item) is not very meaningful (and carries many assumptions).
- **Comparing effects of variables on different scales**: to say "the effect of x_1 is stronger than the effect of x_2", we need a common scale.

Option 1 — standardise the coefficient *after* fitting	**Option 2** — standardise the variables *before* fitting
$\beta_1^* = \beta_1 \cdot s_x / s_y$ standardised slope = unstandardised slope × SD(x) / SD(y)	Convert x and y to **z-scores** (mean 0, SD 1): $z_{x_i} = (x_i - \bar{x}) / s_x$ $z_{y_i} = (y_i - \bar{y}) / s_y$ If **both** x and y are standardised, the fitted coefficients are standardised.

`m1 <- lm(score ~ hours, data = test)`
`1.054545 * (sd(test$hours) / sd(test$score))`
→ **0.721**
	`test <- test |> mutate(`
	`  score_z = (score - mean(score)) / sd(score),`
	`  hours_z = (hours - mean(hours)) / sd(hours))`
	`m2 <- lm(score_z ~ hours_z, data = test)` → slope **0.721**, intercept **0**



Same data, same fit: standardising only rescales the axes. The z-scored line passes through (0, 0).

5. Interpreting standardised coefficients

	Unstandardised	Standardised
Intercept	0.400	0 (shown as 1.17e-17 — rounding error)
Slope	1.0545	0.7212
Residual standard error	1.626	0.7348
t value / p-value of slope	2.945 / 0.0186	**2.945 / 0.0186 — identical**
R^2 / adjusted R^2 / F-test	0.5201 / 0.4601 / 8.67	**identical**

- **Intercept = 0 always** when all variables are standardised, because all means become 0: $\beta_0 = \bar{y} - \beta_1\bar{x} = 0 - \beta_1 \cdot 0 = 0$.
- **Slope**: the change in y **in SD units** for every **1 SD increase** in x. Here: *for every standard deviation increase in hours of study, test score increases by 0.72 standard deviations.*
- Standardising changes the **scale**, not the **evidence**: t, p, R^2 and F are unchanged.

6. Which should we use?

Unstandardised	Standardised
When variables are on **meaningful scales** (e.g. X extra hours of exercise per week adds Y years of healthy life).	When scales are **arbitrary**, or when we want to **compare effects** of variables measured on different scales.

Cautions

- Putting coefficients on a common metric **does not guarantee they can be meaningfully compared**.
- The SD is a poor measure of spread for **skewed** distributions — be cautious standardising skewed variables.

7. Relationship to correlation (r)

With **a single continuous predictor**, the standardised slope β_1^* is **exactly equal to Pearson's r** between predictor and outcome: `lm(score_z ~ hours_z)` gives 0.72 and `cor(test$hours, test$score)` gives 0.72.

- r is a standardised measure of linear **association**; β_1^* is a standardised measure of the linear **slope**.
- With **multiple predictors**, standardised slopes correspond to the **part (semi-partial) correlation** coefficients.
- Bonus check: $R^2 = 0.5201 = r^2 = 0.721^2$ (true for simple regression only).

8. Multiple regression: why add predictors?

The aim of a linear model is to **explain variance in an outcome**. A model can include (in principle) any number of predictors. Predictors may be **correlated with each other**; multiple regression finds the optimal prediction **taking their redundancy into account**.

Use	Idea
Prediction	Several predictors may predict the outcome better than one.
Theory testing	Theories often say several variables together contribute to an outcome.
Covariate control	Assess one predictor's effect while controlling for others (e.g. personality → health, removing age and gender).

9. The multiple regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_j x_{ji} + \varepsilon_i$$

- One extra β for each extra x. Geometrically: 1 predictor → a **line**; 2 predictors → a **plane** in 3D.
- β_0 = predicted y when **all** x's are 0.
- Each β_j is a **partial regression coefficient**: the change in y for a one-unit change in x_j **when all other x's are held constant**.
- "Holding constant" = the effect of a predictor with the other predictors' values **fixed**; lm isolates each predictor's **unique contribution**. Synonyms: **controlling for** (used most in the course), adjusting for, partialling out, residualising for.

Example. 150 students, Biology MCQ exam (max 40). Predictors: hours studied and school motivation (**already standardised**: mean 0, negative = low, positive = high). $Score_i = \beta_0 + \beta_1 \cdot Hours_i + \beta_2 \cdot Motivation_i + \varepsilon_i$. Predictors are separated by +: `lm(score ~ hours + motivation, data = test_study2)`.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.86679    0.65473  10.488   <2e-16 ***
hours        1.37570    0.07989  17.220   <2e-16 ***
motivation   0.91634    0.38376   2.388   0.0182 *
Residual standard error: 4.386 on 147 degrees of freedom
Multiple R-squared: 0.6696,     Adjusted R-squared: 0.6651
```

Coefficient	Interpretation (learn the wording!)
Intercept = 6.87	A student who did **not study** (hours = 0) and has **average motivation** (motivation = 0, since it is standardised) is expected to score **6.87**.
hours = 1.38	**Controlling for motivation**, every additional hour studied is associated with a **1.38-point** increase in test score.
motivation = 0.92	**Controlling for hours of study**, every **1 SD** increase in motivation is associated with a **0.92-point** increase in test score.

Note the mixed units: hours is in raw units (hours), motivation in SD units, the outcome in test points. df = n − (number of β's) = 150 − 3 = **147**.

Using the model for prediction (reconstructing student ID101)

Data: ID101 has score = 7, hours = 2, motivation = −1.42; residual ε = −1.32 (`residuals(m3)[1]`).
Score = 6.87 + (1.38 × 2) + (0.92 × −1.42) + (−1.32) = 6.87 + 2.76 − 1.31 − 1.32 = **7** ✓
Predicted $\hat{y}$ (without ε) = 8.32; residual = 7 − 8.32 ≈ −1.32.

Must remember

- Intercept = expected Y when X = 0 (all X's = 0 in multiple regression) — check whether 0 is meaningful.
- Slope = average change in Y per one-unit change in X, in the variables' units.
- Standardised slope $\beta^* = \beta \cdot s_x / s_y$: change in SD of Y per SD of X; intercept becomes 0; t, p, R^2 unchanged.
- One predictor: β* = r. Several predictors: β* relates to the part correlation.
- Multiple regression: each slope is interpreted **controlling for (holding constant) all other predictors**.